SC7 Bayes Methods MT25

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Lecture 4: Rewards, decisions and utility

Probability as preference

Student: *What is subjective probability?*

Teacher: It expresses the strengths of preferences you hold for different sets of outcomes.

Student: *What is preference?*

Teacher: Your order $A \succ B$ express the belief that $\theta \in A$ is more likely to hold than $\theta \in B$.

Student: *What do you mean by "more likely to hold"?*

Teacher: It's subjectively more probable.

Student: *Aren't we trying to define subjective probability?*

The Axioms of Probability just define algebraic properties of a probability space.

The Frequentist definition of probability (proportion of successes in an infinite sequence of trials) shows us that a data distribution $p^*(\cdot)$ exists for our problem.

It is our job to make an observation model $p(\cdot|\theta)$ that satisfies the AoP and approximates p^*.

The Savage Axioms just define the algebraic properties of preference relations. If preference relations satisfy the SA then a unique prior satisfying the AoP exists.

de Finetti's Theorem shows us that a generative model $\psi \sim F,\ y \sim p^*(\cdot|\psi)$ exists for our problem.

It is our job to write down a set of preference relations satisfying the SA and giving $\pi(\cdot)$ approximating F.

Decision Theory in the context of parameter estimation

SA ensure the whole framework of decision theory is available.

For action δ and truth $\theta \in \Omega$ our loss function is $L(\theta, \delta)$.

The loss is elicited, like the prior. We ask the scientist what they are trying to achieve. Suppose it is parameter estimation.

For data $y \sim p(y|\theta)$ $y \in \mathcal{Y}$ and estimator $\hat{\theta} = \delta(y)$, $\delta : \mathcal{Y} \to \mathcal{D}$ is an action for each $y \in \mathcal{Y}$, with risk

$$\mathcal{R}(\theta, \delta) = \int_{\mathcal{Y}} L(\theta, \delta(y)) p(y|\theta) dy.$$

If we have a prior $\pi(\theta)$, posterior $\pi(\theta|y)$ and marginal likelihood $p(y)$ the expected posterior loss is

$$\rho(\pi, \delta|y) = \int_{\Omega} L(\theta, \delta(y)) \pi(\theta|y) d\theta.$$

The Bayes risk $\rho(\pi, \delta) = E_{\theta \sim \pi}(\mathcal{R}(\theta, \delta))$ is closely related,

$$\rho(\pi, \delta) = \int_\Omega \int_\mathcal{Y} L(\theta, \delta(y)) p(y|\theta) \pi(\theta) dy d\theta,$$

since if $p(y)$ is the marginal likelihood, then $\rho(\pi, \delta)$ is

$$\rho(\pi, \delta) = \int_\mathcal{Y} \rho(\pi, \delta(y)|y) p(y) dy.$$

The Bayes estimator δ^π for θ,

$$\delta^\pi = \arg\min_\delta \rho(\pi, \delta)$$

is given for every $y \in \mathcal{Y}$ by

$$\delta^\pi(y) = \arg\min_\delta \rho(\pi, \delta|y).$$

Example: if $\theta \in \Omega$ is discrete and $L(\theta, \delta) = \mathbb{I}_{\theta \neq \delta}$ so we seek the true θ-value, and everything else is equally bad, then $\rho(\pi, \delta|y) = 1 - \pi(\delta|y)$ and our estimator $\delta(y)$ is the MAP.

Actions, rewards, utility and loss:

Let $r(\theta, \delta)$ be the reward for action δ and truth θ.

Let $U(r)$, $r_{\min} \leq r \leq r_{\max}$ denote the utility of reward r.

Utility is the opposite of loss. In terms of our previous notation,

$$L(\theta, \delta) = \text{const} - U(r(\theta, \delta)),$$

replacing one function L by two, U and r, which must be elicited.

At fixed $y \in \mathcal{Y}$ and $\theta \sim \pi(\cdot|y)$ the random reward $R = r(\theta, \delta(y))$ has distribution $R \sim P_{\delta,y}(\cdot)$.

Changing action δ changes $P_{\delta,y}$. Choose δ to maximise $E(U(R))$:

$$U(R) \sim U(r(\theta, \delta(y))), \quad \theta \sim \pi(\cdot|y)$$

$$E_{\theta|y}(U(r(\theta, \delta(y)))) = \text{const} - E_{\theta|y}(L(\theta, \delta(y))),$$

so maximising expected utility $E(U(R))$ minimises Bayes risk.

Coherent prior belief

Consider two sets, $A, B \subseteq \Omega$, action space $\delta \in \{A, B\}$ reward $r(\theta, \delta) = \mathbb{I}_{\theta \in \delta}$ and utility $U(0) = 0,\ U(1) = u$ for $u > 0$.

The random reward $R \sim P_\delta$ is defined by $R = r(\theta, \delta),\ \theta \sim \pi(\cdot)$.

The reward distribution, $P_\delta = (P_\delta(0), P_\delta(1))$, for $\delta = B$ is

$$P_\delta = (1 - \pi(B), \pi(B))$$

The expected utility $E_\theta(U(r(\theta, \delta)))$ of the action $\delta = B$ is

$$P_\delta(0)U(0) + P_\delta(1)U(1) = u\pi(B)$$

so we maximise $E(U)$ by choosing B if $\pi(B) > \pi(A)$ etc.

For this to work in general we need the prior and utility to exist.

Let $\pi(A) > \pi(B) \Rightarrow A \succ B$ define a preference order $(\mathcal{B}_\Omega, \succeq)$.*

The idea here is that we assume a true generative process

$$\psi \sim F,\ y \sim p^*(\cdot|\psi)$$

exists (for example the data are an IES).

Our preferences $(\mathcal{B}_\Omega, \succeq)$ *are our prior*.

In our model for the true generative process $\psi \in \Omega$ and if $A \succ B$ then we think $F(A) > F(B)$ so we want $\pi(A) > \pi(B)$.

If we start with $(\mathcal{B}_\Omega, \succeq)$ then we seek a prior expressing it,

$$\pi(A) > \pi(B) \quad \Leftarrow \quad A \succ B.$$

If this prior exists and is unique we have coherent belief.

*$(\mathcal{B}_\Omega, \succeq)$ is a binary relation over $A, B \in \mathcal{B}_\Omega$ with $A \sim A$ and if $A \succeq B$ then $A \succ B$ or $A \sim B$. Also, $A \succ B \Leftrightarrow B \prec A$.

Coherent posterior belief

We ultimately need a posterior $\pi(\cdot|y)$ to exist for each $y \in \mathcal{Y}$.

We can then go forward with the reward distribution $R \sim P_{\delta,y}$ defined by $R = r(\theta, \delta(y)),\ \theta \sim \pi(\cdot|y)$.

However, we simply assume the observation model $p(\cdot|\theta)$ exists. If the prior exists and $p(y) < \infty$ then the posterior

$$\pi(\theta|y) = \frac{p(y|\theta)\pi(\theta)}{p(y)}$$

is proper and clearly exists.

This time we don't start with a set of preferences, we get them from the posterior

$$\pi(A|y) > \pi(B|y) \quad \Rightarrow \quad A \succ B.$$

So we focus on existence of the prior.

The Ellsberg paradox This paradox shows that peoples' preferences are in some cases inconsistent with any prior.

Two urns, (A) has 50 red and 50 black balls, (B) has 100, numbers unknown. Bet £1000 on the color of draw. For each bet pick an option.

	Option 1 color$_{\text{urn}}$	Option 2 color$_{\text{urn}}$
Bet 1	r_A	b_A
Bet 2	r_B	b_B
Bet 3	r_A	b_B
Bet 4	r_B	b_A

We are indifferent on Bets 1/2 as "red" and "black" are exchangeable here.

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We are indifferent on Bets 1/2 as "red" and "black" are exchangeable here. However many people prefer fixed odds (urn A) to subjective uncertainty (urn B), taking r_A/b_A in Bets 3/4.

Our preferences for each bet tell us about our unstated prior $\pi(\phi)$ for the probability $\phi = \Pr(b_B)$ to draw black from urn B.

Suppose rewards have utilities $U(-1000) = 0$ and $U(1000) = u$, with $u > 0$. Indifference on Bet 1 is reasonable as

$$E(U|\text{choose } r_A) = E(U|\text{choose } b_A) = u/2.$$

On Bet 2, $E(U|\text{choose } r_B) = u(1 - E_\pi(\phi))$ and

$$E(U|\text{choose } b_B) = uE_\pi(\phi),$$

so indifference here implies $E_\pi(\phi) = 1/2$ in our prior.

On Bet 3, choosing r_A over b_B gives $E_\pi(\phi) < 1/2$. Choosing b_A over r_B in Bet 4 gives $E_\pi(\phi) > 1/2$, a contradiction and so a prior does not exist.

Expected utility hypothesis

Assume coherent prior belief, so $r(\theta, \delta) \sim P_\delta$ exists for $\delta \in \mathcal{D}$.

Consider the order $\succeq_U$ on P_δ and $P_{\delta'}$ given by

$$P_\delta \succ_U P_{\delta'} \quad \Leftarrow \quad E_{P_\delta}(U(R)) > E_{P_{\delta'}}(U(R))$$

If we *start* with a preference over reward distributions then we need a utility function $U(r)$ expressing the order

$$P_\delta \succ_U P_{\delta'} \quad \Rightarrow \quad E_{P_\delta}(U(R)) > E_{P_{\delta'}}(U(R))$$

in order to have coherent inference.

The expected utility hypothesis (EUH) says there exists a prior π and a utility function U such that our order over actions corresponds to order by expected utility.

Coherent inference chooses the action to maximise the expected utility. It is possible if the EUH holds.

The Allais paradox Our preferences may be inconsistent with any utility function. In this example the reward distributions are given, so the "prior" exists.

Let $p = (p_1, p_2, p_3)$ be the probability you win respectively

$$(r_1, r_2, r_3) = (\pounds 0, \pounds 500,000, \pounds 750,000).$$

In each of two rounds you have a choice between two lotteries.

1. (A) with $p^{(A)} = (0, 1, 0)$ *OR* (B) with $p^{(B)} = (0.01, 0.89, 0.1)$
2. (C) with $p^{(C)} = (0.89, 0.11, 0)$ *OR* (D) with $p^{(D)} = (0.9, 0, 0.1)$

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People comonly choose (A) and (D). Choices (B) and (D) maximise the expected return. What utility function are they using?

Set the utilities to be $U(r_1) = 0$, $U(r_2) = u$ and $U(r_3) = 1$.

The expected utilities are $E(U) = (p_1, p_2, p_3)(0, u, 1)^T$, so

$$\begin{aligned} E(U|A) &= u & \qquad E(U|B) &= 0.1 + 0.89u \\ E(U|C) &= 0.11u & \qquad E(U|D) &= 0.1 \end{aligned}$$

Preferring (A) to (B) means

$$u > 0.1 + 0.89u \quad \Rightarrow \quad u > 10/11.$$

On the other hand preferring (D) to (C) means

$$0.1 > 0.11u \quad \Rightarrow \quad u < 10/11,$$

which is a contradiction. This paradox shows that human decision making does not always maximise an expected utility. This is unsurprising. However, choosing (A),(D) does seem reasonable.